DunesForge - Getting Started in 10 minutes

Version 0.85 beta. This page gets you from installation to your first moving dune field. For the full guide open the QuickStart; if something does not work, open Troubleshooting.

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1. Install (2 minutes)

1. Start **Blender 4.2 or newer**.
2. Open **Edit > Preferences > Get Extensions**, click the small arrow in the top-right corner and choose **Install from Disk**. Pick the DunesForge ZIP file. Do not unpack it first.
3. Close **Preferences**. In the 3D Viewport press **N** and click the **DunesForge** tab on the right.
4. The top of the panel shows the version, for example **DunesForge build 0.85.0 beta**.

Updating from an earlier version? Install the new ZIP the same way, then **close and reopen Blender**. Blender keeps the old code in memory until it restarts.

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2. Turn on the GPU (optional, 5 minutes)

DunesForge runs on any computer, but an **NVIDIA** graphics card makes it many times faster.

1. In the DunesForge panel click **Install CuPy GPU Backend**. Internet access must be allowed in **Edit > Preferences > System > Network > Allow Online Access**.
2. Wait until the message says the installation finished. It downloads about 1 to 2 GB.
3. Restart Blender. The panel should now say **GPU Acceleration: ACTIVE**.

No NVIDIA card, a Mac, or no internet? Skip this step. Everything works on the CPU, only slower. Keep the **Resolution** at 256 or lower.

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3. Look at a ready-made scene (1 minute)

1. In **Start with a Demo or Preset** choose **Desert: Dunes + Obstacles**.
2. Click **Create Demo Scene** and wait. The add-on calculates a short warm-up so the dunes already exist when the scene opens.
3. Switch the viewport to **Material Preview** or **Rendered**.
4. Click **Play / Pause**. The dunes start to move with the wind. Click it again to stop.

Every demo has a card in the corner of the camera view that says what it shows and what to try. There are more than 70 demos: snow, water, rain, plants, avalanches and more.

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4. Your own dune field (2 minutes)

1. Open a new scene (**File > New > General**) and delete the default cube.
2. In the DunesForge panel click **Create Terrain**.
3. Under **2. Starting sand**, set **Even sand layer (start)** to about **1 m**.
4. Click **Initialize Simulation**, then **Play / Pause**.
5. To get there faster without playback, set the number next to **Run Steps** (for example 500) and click **Run Steps**. The viewport updates once at the end.
6. Click **Add Surface Shader** under **3. Surface appearance** to give the sand its material.

Wind: Wind Direction and **Wind Strength** are in the **Wind** panel. Change one of them, run more steps and watch how the dunes turn.

Thick sand (1 m and more) forms real dunes with the default settings. If you turn on both **Crest Shear Ceiling** and **Saltation Feedback (Owen)**, it can break into flat steps along the wind; a red

box near the sand layer then points at the fix.

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5. Water (1 minute)

Choose one of the water demos, for example **Water: A Pond Among Dunes (Experimental)**, **Water: Waterfall From A Lake (Experimental)** or **Water: Flash Flood In A Wadi (Experimental)**, click **Create Demo Scene** and then **Play / Pause**. The water flows over the terrain and can move sand.

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6. Save your work

- **File > Save** keeps the scene and all settings.
- The running simulation itself is kept only while Blender is open. To come back to the same state later, use **Save Checkpoint** in the **Export / Bake** panel, or bake the animation.

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7. Things worth knowing

- **Simple, Standard, Advanced:** the **Interface** switch at the top of the panel shows more or fewer controls. A new scene opens in **Simple**. Nothing is lost when you switch; controls are only hidden.
- **Ctrl+Z does not undo the simulation.** Use the DunesForge **Undo** button (or **Ctrl+Alt+Z**) to step the sand back.
- **Help for any control:** right-click it and choose **Explain this DunesForge option**.
- **Start small.** Try ideas at a low **Resolution** first, then raise it for the final result.

Reporting a problem in the beta

Please tell us:

1. The version shown at the top of the panel (for example **0.85.0 beta**) and your Blender version.
2. Your graphics card, and whether the panel says **GPU Acceleration: ACTIVE**.
3. What you did, step by step, what you expected and what happened instead.
4. A screenshot, and the log file: click **Open Log Folder** and attach **tydunes.log**.

A small scene that shows the problem helps more than a long description.